



GEEM 433

A Practical Ethics Toolkit

Ethical Approaches

ENGINEERING ETHICS AND
MORALE PHILOSOPHY

3 Ethical Approaches –

1. Utilitarian Approach
 - a) Cost-benefit Test
 - b) Test of Maximizing Goods
 - c) Applying the Rules and Practices Test
2. The Respect for Persons Approach
 - a) The Golden Rule Test
 - b) The Self-Defeating Test
 - c) The Rights Test
3. The Virtue Ethics Approach

Utilitarian Approach

The fundamental principle of the utilitarian model of common morality is *We should maximize overall well-being. We refer to the population over which well-being is maximized as the audience.*

- To implement the utilitarian approach, we must determine the scope of this audience.
- Once we determine the audience, we must know which course of action will produce the most good in both the short and the long term.

Audience:

Ideally, perhaps the audience should include all humans, or at least all humans who might be affected by the action to be evaluated.

Some utilitarians think even those animals clearly able to experience pain or pleasure should be included in the audience, if they would also be affected.

But then it becomes enormously difficult to calculate which actions produce the most good for so large an audience.

If we limit the audience so that it includes only our country, company, or community, then we face the criticism that others have been arbitrarily excluded. **Therefore, in practice, those with utilitarian sympathies need to develop acceptable ways of limiting the audience**

Cost Benefit Test of Utilitarian Approach

If a utilitarian approach requires that we maximize well-being, how should we go about determining the criteria we should use in seeking this maximization?

One approach that has appeal from the engineering perspective is cost-benefit analysis (CBA), which holds that the course of action that produces the greatest benefit or utility relative to cost should be chosen.

In using this method, one must convert negative and positive utilities into monetary terms. A close relative of CBA is risk-benefit analysis (RBA), which attempts to balance the risk of benefit against the risk of harm. Because it is often more difficult to determine risks than costs, we shall consider only CBA, which involves three steps. Let's look at them.

Applying Cost Benefit Test of Utilitarian Approach

Three (03) Steps:

1. Identify the available options that can provide a solution to the problem under consideration.
2. Assess the costs (measured in monetary terms) and the benefits (also measured in monetary terms) of each option. The costs and benefits must be assessed for the entire audience of the action, or all who are affected by the decision.
3. Make the decision that is likely to result in the greatest benefit relative to cost; that is, the course of action chosen must not be one for which the funds spent on implementing the action could be spent on another action that would better resolve the problem under consideration.

Test of Maximizing Good Consequences

Some utilitarian approaches do not require that values be measured in strictly quantitative terms. However, they do require that we try to determine what will, in some sense, maximize good consequences. Here, we can try to proceed on the assumption that an action is right if it results in more utility than any alternative action that is available in this situation.

In other words, the question is, Will this particular course of action result in more good than any alternative course of action that is available in this situation?

To answer this question, the following procedure is useful.

Applying Test of Maximizing Goods Consequences

1. Identify the available options in this situation.
2. Determine the appropriate audience for the options, keeping in mind the problems in determining the audience.
3. Decide which available option is likely to bring about the greatest good for the appropriate audience, taking into account the harms as well as benefits.

The Rules and Practices Test

It is one thing to ask about the utility of the consequences of a single action, and another thing entirely to ask about the utility of the consequences of a general practice, as this practice is enshrined in a rule or set of rules.

Determining the consequences of a general practice may be more difficult than determining the consequences of a single act, because the number of people affected by a general practice the audience is usually much larger. However, sometimes the consequences of a general practice are so obvious that little imagination is needed to know what the consequences of the policy would be.

The Rules and Practices Test

Think of traffic rules designed to enhance cooperative, safe driving. It is late at night and there seems to be no one around, and the light is red. You might think it is obvious that no one would be harmed and it would be more convenient to you to violate the law and go through the red light.

From a utilitarian perspective, in situations covered by well-understood, generally observed rules or practices that serve utilitarian ends, a case can be made for justifying your actions by appealing directly to these rules and practices. These generally observed rules and practices, in turn, are justified by their utility. In the vast majority of cases, you should probably just abide by the general rules and not even consider whether their violation in a particular case should be justified.

Applying The Rules and Practices Test

1. Identify the established practice, if any, that applies to the appropriate audience in this situation. If the practice promotes utility better than any alternative practice, it should be followed.
2. If there is no applicable practice, select the one whose support in this situation is likely to have the best long-run consequences, all things considered.
3. Follow the justified practice in this situation, unless you think this might be a situation in which exception can justifiably be made on utilitarian grounds



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Welcome!
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Respect for Persons Test

The fundamental principle of the RP model of common morality is “to Act so that you respect all humans as free and equal moral agents”. This equal regard for moral agents can be understood as a basic requirement of justice.

- A moral agent must be distinguished from knives or airplanes, which can only fulfill goals or purposes that are imposed upon them from the outside. Inanimate objects cannot evaluate actions from a moral standpoint.
- A paradigm example of a moral agent is a normal adult human being who, in contrast to inanimate objects, can formulate goals or purposes of his or her own. Such a being is said to have autonomy.
- From the RP standpoint, maximizing the welfare of the majority, as utilitarianism suggests, must take second place to the goal of respecting the moral agency of all individuals. People may not be killed, deceived, denied their freedom, or otherwise violated simply to bring about a greater total amount of utility. As with our treatment of utilitarian thinking, we consider three approaches to RP thinking.

The Golden Rule Test

Reversibility is a special application of the Universalization Principle because the idea of universalization implies that a judgment should not change simply because the roles are reversed.

In thinking about treating others as I would have them treat me, I need to ask what I would think if I were in their position. If I am tempted to tell a lie to escape a particular difficulty, then I need to ask what I would think if the lie were told to me. Universalizing our thinking by applying the idea of reversibility can help us realize that we may be endorsing treating others in ways that we would object to if done to us.

This is the basic idea behind the Golden Rule, Its most familiar formulation in our culture *is* “Do unto others as you would have them do unto you”.

Applying the Golden Rule Test

1. Identify the action that is to be tested by applying the Golden Rule.
2. Ask whether you would be willing to have a similar action done when you are the recipient of the action, assuming that your values and those of others are similar.
3. If you are willing to be the recipient of your contemplated action, the action is morally permissible by the Golden Rule

The Golden Rule does not by itself provide all the criteria that must be met to satisfy the RP standard, but its requirements of universalizability and reversibility are vital steps in satisfying that standard.

The Self-Defeating Test

It is important to realize that using the self-defeating test does not depend on whether anyone actually makes promises without intending to keep them, cheats on exams, or substitutes inferior and cheaper parts in a product.

The question is, What if everyone did this? This is a hypothetical question, not a prediction that others actually will act this way as a result of what someone else does.

Another way of applying the fundamental idea of the Universalization Principle is to ask whether I would be able to perform the action in question if everyone else performed the same action in the same or similar circumstances.

- If everyone else did what I am doing, would this undermine my ability to do the same thing?
- If I must say yes to this question, then I cannot approve others doing the same kind of thing that I have done, and thus universalizing my action would be self-defeating.

To proceed anyway, treating myself as an exception to the rule is to pursue my own good at the expense of others. Thus, it fails to treat them with appropriate respect.

Applying the Self-Defeating Test

1. Identify the action you want to test by the Self-Defeating Test.
2. Ask whether the action would be self-defeating if everyone did it, either because
 - (a) the action could not be performed if everyone did it, or
 - (b) the purpose you have in performing the action would be undermined if everyone did it.
1. If the action fails either (a) or (b), it is impermissible. If not, it is permissible.

The Rights Test

Some theorists in the RP tradition have concluded that one of the most useful ways of formulating the requirement to respect the moral agency of others is to say that we should honor the rights of people that are necessary for them to exercise their moral agency and to pursue their well-being.

A right may be understood as both an entitlement to act and an entitlement to have another individual act in a certain way. Because of this dual aspect, rights are often thought of as existing in a correlative relationship with duties.

Thus, if Kelly has a right to life, others have a duty not to kill Kelly. If Kelly has a right to bodily integrity, others have a duty not to cause bodily harm to Kelly.

Gewirth's Hierarchy of Rights

Tier 1. The most basic rights, the essential preconditions of action: for example, life, physical integrity, and mental health.

Tier 2. Rights to maintain the level of purpose fulfillment one already has, such as the right not to be deceived or cheated, the right to informed consent to unusual risks, the right not to have possessions stolen, the right not to be defamed, and the right not to suffer broken promises.

Tier 3. The rights necessary to increase one's level of purpose fulfillment: for example, the right to attempt to acquire property and wealth

Applying the Rights Test

1. Identify the action or rule to be evaluated and the available options.
2. Determine what options are available and what rights are at stake in each of the options.
3. Determine the place in the hierarchy of rights at stake and whether the rights are violated or infringed.
4. Identify the action or rule that will produce the least serious violations or infringements of the most significant rights.
5. Make a choice that seems likely to produce the least serious rights infringements or violations, all things considered.

Applying the Rights Test

Using this hierarchy, it would be wrong for a plant manager to attempt to save money by emitting a pollutant that is highly carcinogenic, because the right to life is a first-tier right and the right to acquire and use property and wealth for one's benefit is a third-tier right.

Sometimes, however, the hierarchy is more difficult to apply. How shall we balance a slight infringement of a first-tier right against a much more serious infringement or outright violation of a second-tier or third-tier right?

The hierarchy of rights provides no automatic answer to such questions. Nevertheless, it provides a framework for addressing them

Virtue Ethics

Virtue ethics, perhaps the oldest tradition of ethical thought, has become increasingly important among contemporary ethicists. The fundamental principle of virtue ethics is Act in the way the good or virtuous person would act in the circumstances.

A virtue is usually described as a dispositional trait, that is, a character trait that disposes or inclines a person to do the right thing. A virtue can be described as both deep and wide. It is deep in the sense that a virtue is a firmly entrenched habit that leads a person to consistently act in a certain way and to which he is strongly committed. It is wide in that it manifests itself in a variety of ways. A virtuous person exhibits virtue not only in actions but also in emotional reactions, in interests, and in general sensibilities. A truly honest person is not honest simply because she thinks it is the best way to stay out of trouble, but because she genuinely believes that being honest is the best way to live. She is disgusted by people who are dishonest and does not enjoy being in their company.

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Virtue Ethics

In order to better understand what a virtue is, virtue ethicists such as Aristotle have found it useful to think of virtues as occupying a middle position (or mean) between vices. We can think of courage as a middle ground between the vice of cowardice on the one hand and the vice of foolhardiness on the other. We can think of the virtue of generosity as a middle ground between the vice of miserliness on the one hand and the vice of being a spendthrift on the other. We can think of the virtue of loyalty to an employer as a middle ground between the vice of complete disloyalty on the one hand and the vice of unquestioning obedience to the employer on the other.¹¹

Core Virtue & Selected Character Strengths

1. Wisdom (creativity, open-mindedness, perspective)
2. Courage (bravery, persistence, vigor or energy)
3. Humanity (love, kindness)
4. Justice (citizenship, fairness, leadership)
5. Temperance (modesty, self-control)
6. Transcendence (appreciation of beauty and excellence, gratitude, spirituality)

Applying the Virtue Ethics

1. Determine the alternative courses of action that are possible in the situation.
2. Determine the virtues (or vices) that correspond to these courses of action.
3. Evaluate the actions in terms of the virtues (or vices) that motivate them.
 1. If a course of action is motivated by vices, a course of action motivated by virtues should be chosen instead.
 2. If two or more courses of action are motivated by different virtues, the course of action motivated by the most appropriate virtues for the situation should be chosen.
4. If no decision can be made as to which virtues are most appropriate, the actions associated with different virtues are equally permissible.

Professional Care

Engineers recognize the importance of this virtue, because they often refer to taking due care as important for members of their profession.

- Care is a disposition to both protect and promote the well-being of another in the case of engineering, the wellbeing of the public.
- The paradigm of care is the relationship of parents to their children. **The care relationship to children has two dimensions:** protecting children from harm, and promoting their well-being.
- In manifesting the virtue of care for the public, these same dimensions are important. Engineers must insure that they not only do not harm the health, safety, and welfare of the public (**prohibitive and preventive ethics**), but also promote the well-being of the public through their professional work (**aspirational ethics**).
- As an optional further extension of aspirational ethics, engineers may devote themselves to improving the well-being of the poor and disadvantaged.

Virtue Ethics: An Application

Because applying virtue ethics may cause special difficulties, we supply an example of how these guidelines apply to a moral issue.

"After completing his degree in chemical engineering in June, Gerald is scheduled to return to the family farm to help with its operation. In early May, his father became seriously ill, and Gerald is convinced that the only way to save the family farm is to take a job in engineering. Most of his fellow seniors have already taken jobs, and the interviewing season is over. The only employment opportunity Gerald finds is with Pro-Growth Pesticides. The family farm is an organic farm, however, and Gerald's father has always strongly opposed the use of pesticides. Gerald himself has become convinced that pesticides harm the environment generally and farm products in particular. He knows that he will be asked about his views on pesticides."

What should he do? Let us look at the issue from the standpoint of virtue ethics.

Virtue Ethics: An Application

1. Gerald must first determine the alternative actions that are possible in the situation. He thinks of three possibilities. He could:
 - (a) refuse to interview for the job,
 - (b) interview but answer questions about pesticides honestly, or
 - (c) interview and misrepresent his views on pesticides in order to get the job.

Virtue Ethics: An Application

2. In determining the virtues and vices that correspond to these courses of action, Gerald might come up with the following analysis.

(a) If he refuses to interview for the job, he will continue to be an honest person and also maintain his integrity. That is, he will continue to be a person who acts consistently with his principles. He may, however, not manifest proper loyalty to his family members, because he would fail to help them keep the farm. He would show loyalty to the family ideals, however.

(b) If he interviews for the job but answers questions about his views on pesticides honestly, he will preserve his honesty, but may compromise his integrity by applying for (and possibly getting) a job that would be inconsistent with his principles. He will preserve his loyalty to his family, however, at least in the sense of helping to keep the farm.

(c) If he interviews for the job and misrepresents his views on pesticides, he will manifest loyalty to his family, but fail to be honest or a person of integrity.

Virtue Ethics: An Application

3. Now we must evaluate the alternatives open to Gerald in terms of whether they are grounded, either directly or indirectly, in the virtues appropriate to a morally worthy person.

How should he evaluate these three options? He will fail to properly manifest at least one of the virtues he prizes no matter what he does.

- Options (a) and (b) will violate only one of the virtues, while (c) will violate two.
 - Option (c), furthermore, seems to more directly manifest the vices of dishonesty and lack of integrity.
- While (c) seems to be the least desirable choice, the choice between (a) and (b) is more difficult.
 - If he can find another job opportunity, clearly option (a) is the most desirable.
 - If this possibility is not open to him, it may depend upon what kind of person Gerald most wants to be: a loyal person or a person with integrity.

Virtue Ethics: An Application

3.

Gerald may well decide that he wants to manifest integrity by being consistent with his own values even more than he wants to be loyal to his family, so that he should refuse the interview.

Gerald may be able to find a creative middle way, so that such a difficult choice will not have to be made.